OPERATING SYSTEMS FINAL PROJECT

MiniOS Omega (1063)

User Manual

Version 1.0 | A command-line operating system simulator written in Python, run from Windows Command Prompt (CMD).

Contents

1. Introduction

2. System Requirements

3. Installation

4. How to Start MiniOS

5. Login Instructions

6. Command List

7. Common Errors

8. How to Exit MiniOS

1. Introduction

MiniOS Omega (1063) is a simple command-line operating system *simulator*. It is a Python program that behaves like a small operating system: it has a login screen, its own command prompt, a virtual file system, simulated processes, simulated memory, and system information.

It is not a real bootable operating system. Everything it shows - files, processes, memory and system details - is simulated inside the program and does not touch your real Windows files, programs, or RAM.

With MiniOS you can:

Area	What you can do
Command line	Type commands at the `MiniOS>` prompt and get help, date, and time
Users	Log in as **admin** or **student** and check who is logged in
Files	Create folders and files, write and read text, move between folders, delete
Processes	Start simulated programs, list them, and terminate them
Memory	View, allocate, and release simulated memory (out of 1024 MB)
System	View simulated system information

Important: MiniOS keeps everything in memory only. When you exit, all files, folders, processes, and allocated memory are discarded and the next start begins fresh.

2. System Requirements

Requirement	Details
Operating system	Windows (run through Command Prompt / CMD)
Software	Python 3.x installed and added to PATH
Hardware	Any computer that runs Python; no special hardware or graphics needed
Extra packages	None. MiniOS uses only Python's standard library
Interface	Keyboard only. There is no graphical interface

To check that Python is installed, open CMD and type `python --version`. A version number such as `Python 3.12.0` should appear.

3. Installation

MiniOS does not need an installer. Simply place the project folder on your computer.

Project files

File	Purpose
main.py	Starts the system; command prompt and general commands
users.py	User accounts and login
filesystem.py	File and folder commands
processes.py	Process commands
memory.py	Memory commands

Steps

1. Download or clone the MiniOS repository, or extract the MiniOS ZIP file.
2. Make sure all five `.py` files are in the **same folder**.
3. Confirm Python is installed (see Section 2).
4. Open Command Prompt and go to the project folder, for example:

```
cd C:\Users\YourName\Downloads\MiniOS
```

4. How to Start MiniOS

In CMD, inside the MiniOS folder (the one containing `main.py`), run:

```
python main.py
```

The login screen appears:

```
========================================
             MiniOS Login
========================================

Username:
```

5. Login Instructions

MiniOS has two built-in accounts:

Username	Password	Role
admin	admin123	Administrator
student	student123	Standard User

Steps

1. Type your username and press Enter. Usernames are case-sensitive.

2. Type your password and press Enter. **The password is hidden** as you type, so nothing appears on screen; this is normal.

3. If correct, MiniOS greets you and shows the main banner and prompt.

```
Login successful.

Welcome, admin!

========================================
        MINIOS OMEGA(1063) V1.0
========================================

Welcome to MiniOS!

MiniOS>
```

Failed logins: You have **3 attempts**. After each wrong attempt MiniOS shows how many remain. After the third failure the system locks and the program ends; run `python main.py` again to retry.

```
Error: Invalid username or password. Attempts left: 2
```

6. Command List

Type a command at the `MiniOS>` prompt and press Enter. Command names are **not** case-sensitive (`HELP` works), but file, folder and program names are. Arguments are separated by spaces, so names cannot contain spaces. Blank lines are ignored.

Quick reference

Command	Purpose	Syntax
help	Show all commands	help
clear	Clear the screen	clear
date	Show current date	date
time	Show current time	time
whoami	Show current user	whoami
systeminfo	Show system information	systeminfo
exit	Shut down MiniOS	exit
dir	List files and folders	dir
mkdir	Create a directory	mkdir <folder>
create	Create an empty file	create <file>
write	Write text to a file	write <file> <text>
read	Display a file	read <file>
delete	Delete a file or empty folder	delete <name>
cd	Change directory	cd <folder> | cd ..
process	List processes	process
run	Start a process	run <program>
kill	Terminate a process	kill <pid>
memory	Show memory status	memory
allocate	Allocate memory (MB)	allocate <MB>
free	Release memory (MB)	free <MB>

6.1 General Commands

COMMAND: help

Purpose: Displays every available command with a short description.

Syntax:

```
help
```

Example:

```
MiniOS> help

Available Commands:

  help         - Show available commands
  clear        - Clear the screen
  date         - Show current date
  ...
```

COMMAND: clear

Purpose: Clears the screen.

Syntax:

```
clear
```

Example:

```
MiniOS> clear
```

COMMAND: date

Purpose: Shows the current date in YYYY-MM-DD format.

Syntax:

```
date
```

Example:

```
MiniOS> date
Date: 2026-10-23
```

COMMAND: time

Purpose: Shows the current time in 24-hour HH:MM:SS format.

Syntax:

```
time
```

Example:

```
MiniOS> time
Time: 15:51:07
```

COMMAND: whoami

Purpose: Shows the user who is currently logged in.

Syntax:

```
whoami
```

Example:

```
MiniOS> whoami
Current User: admin
```

COMMAND: systeminfo

Purpose: Displays simulated system information.

Syntax:

```
systeminfo
```

Example:

```
MiniOS> systeminfo

========================================
          SYSTEM INFORMATION
========================================

OS Name: MiniOS Omega(1063)
Version: 1.0
Architecture: 64-bit
CPU: Virtual CPU
Memory: 1024 MB
User: admin

========================================
```

COMMAND: exit

Purpose: Shuts down MiniOS and returns you to CMD.

Syntax:

```
exit
```

Example:

```
MiniOS> exit

Shutting down MiniOS...

Goodbye!
```

6.2 File Management Commands

Files live in a virtual file system that starts empty. Paths are relative to the current directory; type a single name, not a full path.

COMMAND: dir

Purpose: Lists the files and folders in the current directory. Folders show <DIR>; files show their size in bytes (characters).

Syntax:

```
dir
```

Example:

```
MiniOS> dir

Directory of \

  <DIR>    Documents
  23 B     notes.txt
```

Notes: If the directory has nothing in it, `(empty)` is shown.

COMMAND: mkdir

Purpose: Creates a new directory.

Syntax:

```
mkdir <folder>
```

Example:

```
MiniOS> mkdir Documents
Directory created.
```

Possible errors: `Usage: mkdir <folder>` (no name given); `Error: Name already exists.`

COMMAND: create

Purpose: Creates a new, empty file.

Syntax:

```
create <file>
```

Example:

```
MiniOS> create notes.txt
File created.
```

Possible errors: `Usage: create <file>`; `Error: Name already exists.`

COMMAND: write

Purpose: Saves text into an existing file. The new text **replaces** the old content.

Syntax:

```
write <file> <text>
```

Example:

```
MiniOS> write notes.txt Operating Systems Final Project
File saved.
```

Notes: Create the file first with `create`. Everything after the file name is saved as the text.

Possible errors: `Usage: write <file> <text>`; `Error: File does not exist.`

COMMAND: read

Purpose: Displays the contents of a file.

Syntax:

```
read <file>
```

Example:

```
MiniOS> read notes.txt

Operating Systems Final Project
```

Notes: An empty file shows `(empty file)`.

Possible errors: `Usage: read <file>`; `Error: File does not exist.` (also shown if the name is a folder)

COMMAND: delete

Purpose: Deletes a file or an **empty** directory.

Syntax:

```
delete <name>
```

Example:

```
MiniOS> delete notes.txt
Deleted.
```

Notes: A folder that still contains items must be emptied first.

Possible errors: `Error: File or directory does not exist.`; `Error: Directory is not empty.`

COMMAND: cd

Purpose: Moves into a folder, or back to the parent folder with `..`. The `dir` header shows where you are.

Syntax:

```
cd <folder>
cd ..
```

Example:

```
MiniOS> cd Documents
MiniOS> dir

Directory of \Documents

  (empty)

MiniOS> cd ..
```

Notes: `cd ..` at the top level does nothing. The prompt does not change when you move folders.

Possible errors: `Usage: cd <folder> or cd ..`; `Error: Directory does not exist.`

6.3 Process Management Commands

Processes are simulated. Starting one does not launch a real Windows program. Any name may be used (for example *calculator* or *editor*). Process IDs (PIDs) start at 1 and are never reused in the same session.

COMMAND: run

Purpose: Starts a simulated process and assigns it a PID.

Syntax:

```
run <program>
```

Example:

```
MiniOS> run calculator

Process started.

PID: 1
Name: calculator
Status: RUNNING
```

Notes: Only the first word is used as the program name.

Possible errors: `Usage: run <program>`

COMMAND: process

Purpose: Lists all running processes.

Syntax:

```
process
```

Example:

```
MiniOS> process

PID    NAME           STATUS
1      calculator     RUNNING
2      editor         RUNNING
```

Notes: With none running, `No running processes.` is shown.

COMMAND: kill

Purpose: Terminates a process by its PID and removes it from the list.

Syntax:

```
kill <pid>
```

Example:

```
MiniOS> kill 1
Process 1 terminated.
```

Possible errors: `Usage: kill <pid>`; `Error: PID must be a number.`; `Error: Process does not exist.`

6.4 Memory Management Commands

MiniOS simulates **1024 MB** of memory. **256 MB** is always reserved for MiniOS itself, so a fresh session starts with 256 MB used and 768 MB free. Amounts are whole numbers of megabytes.

COMMAND: memory

Purpose: Shows total, used, and free memory.

Syntax:

```
memory
```

Example:

```
MiniOS> memory

Memory Status
------------------------
Total Memory:  1024 MB
Used Memory:    256 MB
Free Memory:    768 MB
```

COMMAND: allocate

Purpose: Allocates the given number of MB from free memory.

Syntax:

```
allocate <MB>
```

Example:

```
MiniOS> allocate 200
200 MB allocated.
```

Possible errors: `Usage: allocate <MB>`; `Error: Amount must be a positive number.`; `Error: Not enough free memory.`

COMMAND: free

Purpose: Releases memory you previously allocated. The 256 MB used by MiniOS cannot be released.

Syntax:

```
free <MB>
```

Example:

```
MiniOS> free 200
200 MB released.
```

Possible errors: `Usage: free <MB>`; `Error: Amount must be a positive number.`; `Error: Cannot free more memory than allocated.`

7. Common Errors

MiniOS does not crash on mistakes. It prints a message and returns to the prompt.

Message	Cause	How to fix
Unknown command. Type 'help' to see the available commands.	Command misspelled or not supported	Check spelling; type `help`
Error: Invalid username or password. Attempts left: N	Wrong login details	Re-enter using the accounts in Section 5. After 3 failures the system locks; restart MiniOS
Usage: ...	Required argument missing	Follow the syntax shown
Error: Name already exists.	A file or folder with that name is already in the current folder	Choose a different name
Error: File does not exist.	No such file (or the name is a folder) in the current folder	Check with `dir`; check spelling and capitals; `cd` to the right folder
Error: File or directory does not exist.	Nothing with that name to delete	Check with `dir`
Error: Directory is not empty.	Deleting a folder that has items	Delete the items inside first
Error: Directory does not exist.	`cd` target missing or is a file	Check with `dir`
Error: PID must be a number.	Non-numeric PID given to `kill`	Use a number from `process`
Error: Process does not exist.	No process with that PID	Run `process` to see valid PIDs
Error: Amount must be a positive number.	Memory amount is not a whole number above 0	Type a whole number such as 100
Error: Not enough free memory.	Requested more than the free memory	Check `memory`; request less or free some
Error: Cannot free more memory than allocated.	Trying to release more than you allocated	Check `memory`; release a smaller amount

Example:

```
MiniOS> hellooo
Unknown command.

Type 'help' to see the available commands.

MiniOS> read missing.txt
Error: File does not exist.

MiniOS> kill 999
Error: Process does not exist.
```

8. How to Exit MiniOS

Type `exit` at the prompt and press Enter:

```
MiniOS> exit

Shutting down MiniOS...

Goodbye!
```

You can also press **Ctrl+C** at the prompt, which performs the same shutdown. You return to the normal CMD prompt. Remember that all files, folders, processes and memory allocations are cleared when MiniOS closes.